Advanced Java Skills: From Language Internals to Production Engineering

1. JVM Architecture & Memory Model

1. Heap (Young Gen, Old Gen), Stack, Metaspace
2. Object allocation and lifecycle
3. Garbage Collection algorithms (G1, ZGC basics)
4. Class loading mechanism (Bootstrap, Extension, Application ClassLoader)
5. Memory leaks and profiling techniques
6. JVM tuning parameters and performance optimization

2. Concurrency & Multithreading Deep Dive

1. Java Memory Model (JMM) and happens-before relationship
2. Thread lifecycle, synchronization primitives
3. Locks: ReentrantLock, ReadWriteLock, StampedLock
4. ExecutorService, ForkJoinPool, CompletableFuture
5. Lock-free programming and CAS operations
6. Handling race conditions, deadlocks, livelocks

3. Data Structures & Performance Engineering

1. Internal working of HashMap, ConcurrentHashMap
2. Tree structures and balancing algorithms
3. Graph traversal algorithms (BFS, DFS)
4. Time/space trade-offs and performance tuning
5. Memory-efficient data structures

4. Java I/O, NIO & Networking

1. Blocking I/O vs Non-blocking I/O
2. NIO Channels, Buffers, Selectors
3. File handling and serialization
4. Socket programming basics
5. Reactive streams and backpressure concepts

5. Functional Programming & Streams

1. Lambda expressions and functional interfaces
2. Streams API and parallel streams
3. Optional and null safety
4. Functional composition and immutability

6. Object-Oriented Design & Patterns

1. SOLID principles and clean architecture
2. Design patterns (Factory, Builder, Strategy, Observer, Singleton)
3. Domain-driven design basics
4. Code modularity and maintainability

7. Exception Handling & Resilience

1. Checked vs unchecked exceptions
2. Custom exception design
3. Retry mechanisms and fallback strategies
4. Circuit breaker patterns (Resilience4j concept)

8. Performance, Profiling & Tuning

1. CPU and memory profiling
2. Garbage collection tuning
3. Thread pool tuning
4. Latency vs throughput trade-offs

9. Secure Coding Practices

1. Input validation and sanitization
2. Serialization vulnerabilities
3. Secure handling of secrets
4. Preventing injection attacks

10. Integration with Modern Systems

1. Java with Spring Boot microservices

2. Event-driven systems (Kafka basics)
3. Caching (Redis)
4. Distributed system considerations (CAP theorem)